

A conditional reduction of the endpoint complex Hardy–BMO problem to the h_∞ and conditioned endpoints

Automatic Math Discovery Run `fa.banach.001`

June 16, 2026

Status

This is a conditional reduction for the open problem in Randrianantoanina’s paper [1]. It does not claim a full solution. It proves that, in the range $r > 1$, the missing reverse inclusion for the endpoint bmo^c complex interpolation problem follows from the corresponding reverse inclusion for the endpoint h_∞^c complex interpolation problem. It also records a further sufficient reduction to an endpoint interpolation statement for Junge’s conditioned sequence spaces. The h_∞^c complex endpoint is explicitly left open in the later paper [3].

Source problem

The source paper proves that, for finite indices $0 < p, q < \infty$,

$$[h_p^c(\mathcal{M}), h_q^c(\mathcal{M})]_\theta = h_r^c(\mathcal{M}), \quad \frac{1}{r} = \frac{1-\theta}{p} + \frac{\theta}{q}.$$

Immediately after that theorem, it asks whether the corresponding statement remains valid when $0 < p < 1$ and the endpoint couple $(h_p^c(\mathcal{M}), bmo^c(\mathcal{M}))$ is used.

From combining (3.12) and (3.13), we obtain the desired equivalence. $\square$

When $0 < p < 1$, we do not know if the corresponding statement to Theorem 3.16 remains valid if the interpolation couple $(h_p^c(\mathcal{M}), bmo^c(\mathcal{M}))$ is used. Since reiteration theorem is not available for complex interpolations of quasi-Banach spaces, in general, this consideration is independent of Theorem 3.16. We leave this as an open problem.

The same method of proofs can be applied to conditioned spaces and spaces of adapted se-

Thus the desired full endpoint identity is

$$[h_p^c(\mathcal{M}), bmo^c(\mathcal{M})]_\theta = h_r^c(\mathcal{M}), \quad \frac{1}{r} = \frac{1-\theta}{p}.$$

Conditional dependencies

The first unproved dependency in the reduction is the following endpoint reverse inclusion.

Conditional Dependency 1 (h_∞^c complex endpoint reverse inclusion). *Let $0 < p < 1$, $0 < \theta < 1$, and $1/r = (1 - \theta)/p$, with $r > 1$. Assume that*

$$h_r^c(\mathcal{M}) \hookrightarrow [h_p^c(\mathcal{M}), h_\infty^c(\mathcal{M})]_\theta$$

continuously.

The full complex identity

$$[h_p^c(\mathcal{M}), h_\infty^c(\mathcal{M})]_\theta = h_r^c(\mathcal{M})$$

would imply this dependency. Randrianantoanina [3] proves the corresponding real/K-method theorem for (h_p^c, h_∞^c) , but notes that the complex version remains open.

This packet also isolates a lower-level sufficient dependency in the conditioned sequence spaces. Let $L_s^{\text{cond}}(\mathcal{M}; \ell_2^c)$ denote Junge's column conditioned space, with the L_∞ -endpoint framework as in [3]. Let

$$\Pi(a) = \sum_{n \geq 1} (\mathcal{E}_n(a_n) - \mathcal{E}_{n-1}(a_n))$$

be the natural martingale-difference projection, initially defined on finite sequences. By the Kadison–Schwarz calculation in [1, Lemma 2.2], Π is contractive on finite conditioned square functions.

Conditional Dependency 2 (conditioned L_∞ endpoint interpolation). *Let $0 < p < 1$, $0 < \theta < 1$, and $1/r = (1 - \theta)/p$, with $r > 1$. Assume that the endpoint conditioned couple is compatible with Π and satisfies*

$$L_r^{\text{cond}}(\mathcal{M}; \ell_2^c) \hookrightarrow [L_p^{\text{cond}}(\mathcal{M}; \ell_2^c), L_\infty^{\text{cond}}(\mathcal{M}; \ell_2^c)]_\theta$$

continuously. Here compatibility means that the finite-sequence projection Π extends boundedly at the L_∞^{cond} endpoint onto $h_\infty^c(\mathcal{M})$, as it does for $0 < s < \infty$ in [1, Lemma 2.2].

Idea of the proof

The previous one-sided packet used Wolff's inclusion lemma to prove

$$[h_p^c(\mathcal{M}), bmo^c(\mathcal{M})]_\theta \hookrightarrow h_r^c(\mathcal{M})$$

for $r > 1$. To get equality, only the reverse inclusion is missing. The stronger endpoint h_∞^c sits continuously inside bmo^c : bounded full conditioned square function controls the conditioned BMO tail norms. Complex interpolation is monotone under continuous embeddings of endpoint couples. Therefore a reverse inclusion through h_∞^c immediately passes to the larger endpoint bmo^c .

The additional conditioned-space reduction uses the same projection Π . If the martingale difference sequence $D_c x = (dx_n)$ of an element $x \in h_r^c$ has an admissible analytic representation in the conditioned couple, applying Π pointwise to that analytic family produces an admissible analytic representation in the Hardy couple and keeps the value $\Pi D_c x = x$.

Lemma 1. *There is a continuous embedding*

$$h_\infty^c(\mathcal{M}) \hookrightarrow bmo^c(\mathcal{M}).$$

Proof. For a finite martingale x , the bmo^c tail seminorm is controlled by the bounded conditioned square function. Indeed, for $m > n$, martingale orthogonality gives

$$\mathcal{E}_n(|x_m - x_n|^2) = \mathcal{E}_n \left(\sum_{k=n+1}^m |dx_k|^2 \right) = \mathcal{E}_n \left(\sum_{k=n+1}^m \mathcal{E}_{k-1}(|dx_k|^2) \right).$$

The last positive operator is dominated in norm by $\|s_c(x)^2\|_\infty$. The first-coordinate term is also controlled by the standard h_∞^c norm convention. Passing from finite martingales to the completed endpoint gives the continuous embedding. $\square$

Theorem 1. *Let $0 < p < 1$, $0 < \theta < 1$, and put*

$$\frac{1}{r} = \frac{1 - \theta}{p}.$$

Assume $r > 1$ and assume the h_∞^c complex endpoint reverse inclusion above. Then

$$[h_p^c(\mathcal{M}), \mathbf{bmo}^c(\mathcal{M})]_\theta = h_r^c(\mathcal{M})$$

with equivalent quasi-norms.

Proof. The previous one-sided result proves

$$[h_p^c(\mathcal{M}), \mathbf{bmo}^c(\mathcal{M})]_\theta \hookrightarrow h_r^c(\mathcal{M})$$

when $r > 1$. For completeness, we recall the mechanism. Choose $\eta \in (0, 1)$ with

$$1 = \frac{1 - \eta}{p} + \frac{\eta}{r}.$$

The finite-index complex interpolation theorem of [1, Theorem 3.16] gives

$$[h_p^c(\mathcal{M}), h_r^c(\mathcal{M})]_\eta = h_1^c(\mathcal{M}).$$

Setting $\phi = 1 - 1/r$, the Banach endpoint theorem of Bekjan–Chen–Perrin–Yin [2, Theorem 4.4], with the couple reversed, gives

$$[h_1^c(\mathcal{M}), \mathbf{bmo}^c(\mathcal{M})]_\phi = h_r^c(\mathcal{M}).$$

Wolff’s inclusion lemma, in the quasi-Banach inclusion form used in [1, Corollary 3.19], then yields

$$[h_p^c(\mathcal{M}), \mathbf{bmo}^c(\mathcal{M})]_\theta \subset h_r^c(\mathcal{M}),$$

after the standard parameter calculation $\theta = \phi/(1 - \eta + \eta\phi)$.

It remains to prove the reverse inclusion under the stated dependency. By the lemma, the identity map gives a continuous endpoint embedding

$$(h_p^c(\mathcal{M}), h_\infty^c(\mathcal{M})) \longrightarrow (h_p^c(\mathcal{M}), \mathbf{bmo}^c(\mathcal{M})).$$

The complex interpolation functor is monotone for such endpoint embeddings, so

$$[h_p^c(\mathcal{M}), h_\infty^c(\mathcal{M})]_\theta \hookrightarrow [h_p^c(\mathcal{M}), \mathbf{bmo}^c(\mathcal{M})]_\theta.$$

Combining this with the conditional dependency gives

$$h_r^c(\mathcal{M}) \hookrightarrow [h_p^c(\mathcal{M}), h_\infty^c(\mathcal{M})]_\theta \hookrightarrow [h_p^c(\mathcal{M}), \mathbf{bmo}^c(\mathcal{M})]_\theta.$$

Together with the first inclusion, this proves the conditional equality. $\square$

Theorem 2 (conditioned endpoint reduction). *Let $0 < p < 1$, $0 < \theta < 1$, and $1/r = (1 - \theta)/p$, with $r > 1$. Under the conditioned L_∞ endpoint interpolation dependency above,*

$$\mathfrak{h}_r^c(\mathcal{M}) \hookrightarrow [\mathfrak{h}_p^c(\mathcal{M}), \mathfrak{h}_\infty^c(\mathcal{M})]_\theta.$$

Consequently,

$$[\mathfrak{h}_p^c(\mathcal{M}), \mathfrak{bmo}^c(\mathcal{M})]_\theta = \mathfrak{h}_r^c(\mathcal{M})$$

with equivalent quasi-norms.

Proof. Let $x \in \mathfrak{h}_r^c(\mathcal{M})$. Its martingale difference sequence $D_c x = (dx_n)$ belongs to $L_r^{\text{cond}}(\mathcal{M}; \ell_2^c)$, with

$$\|D_c x\|_{L_r^{\text{cond}}} = \|x\|_{\mathfrak{h}_r^c}.$$

By the conditioned endpoint dependency, $D_c x$ belongs to

$$[L_p^{\text{cond}}(\mathcal{M}; \ell_2^c), L_\infty^{\text{cond}}(\mathcal{M}; \ell_2^c)]_\theta$$

with controlled norm. Choose an admissible analytic family F in the conditioned couple such that $F(\theta) = D_c x$. Applying the bounded endpoint projection Π pointwise gives an admissible analytic family ΠF in $(\mathfrak{h}_p^c(\mathcal{M}), \mathfrak{h}_\infty^c(\mathcal{M}))$. Its value at θ is

$$(\Pi F)(\theta) = \Pi(D_c x) = x.$$

Thus

$$x \in [\mathfrak{h}_p^c(\mathcal{M}), \mathfrak{h}_\infty^c(\mathcal{M})]_\theta$$

with norm bounded by the conditioned endpoint constant and the endpoint norms of Π . This proves the $\mathfrak{h}_\infty^c$ reverse inclusion. The preceding theorem then gives the $\mathfrak{bmo}^c$ -endpoint equality. $\square$

Verification notes

The proof is symbolic and uses no computation. The first new point beyond the previous one-sided packet is the endpoint monotonicity step. The second new point is the projection reduction from Hardy martingales to conditioned sequences. The $\mathfrak{h}_\infty^c$ dependency is strictly weaker than the full open $\mathfrak{h}_\infty^c$ complex identity, since only the reverse inclusion is needed. The conditioned dependency is a concrete sufficient route to that reverse inclusion.

Limitations and failed closing routes

This packet does not prove the $\mathfrak{h}_\infty^c$ complex endpoint dependency. The 2022 paper [3] proves K-closedness and real interpolation for $(\mathfrak{h}_p^c, \mathfrak{h}_\infty^c)$, then explicitly states that the complex interpolation version remains open. The 2024 quasi-Banach function-space Wolff theorem of Egert–Kosmala [4] is also not directly applicable: it is a theorem for quasi-Banach function spaces with lattice structure, and it explicitly leaves the corresponding general A -convex quasi-Banach-space reiteration problem open.

The natural attempt to prove the conditioned endpoint dependency is to use the Junge embedding

$$U : L_s^{\text{cond}}(\mathcal{M}; \ell_2^c) \rightarrow L_s(\mathcal{M} \overline{\otimes} B(\ell_2(\mathbb{N}^2))).$$

For finite $0 < s < \infty$, the source paper proves the needed complex interpolation theorem. At $s = \infty$, however, the most direct common retraction of the range of U would require a uniform endpoint column Stein estimate for the adapted projection

$$(\xi_n) \mapsto ((\mathcal{E}_{n-1} \otimes \text{id})(\xi_n))$$

on $L_\infty(\mathcal{M}; \ell_2^c)$. This is precisely the endpoint estimate that is absent from noncommutative Stein theory; the source papers only use complementation in the interior range $1 < s < \infty$. Thus the obvious ambient projection route does not close the problem without an additional endpoint hypothesis. For finite filtrations one obtains such a projection with constants depending on the number of levels, but that does not give the uniform theorem asked for in the source problem.

The range $r \leq 1$ is not covered. Even with the h_∞^c reverse inclusion, one would still need an inclusion

$$[h_p^c(\mathcal{M}), \text{bmo}^c(\mathcal{M})]_\theta \hookrightarrow h_r^c(\mathcal{M})$$

in that range, and the current Wolff/Banach-endpoint argument only gives this for $r > 1$.

Novelty check

Local indexes were searched for arXiv:2108.06341, h_p^c , bmo^c , h_∞^c , conditioned spaces, and endpoint complex interpolation. Web and arXiv searches found the source paper [1], the later h_∞^c K-closedness paper [3], Pisier's classical and triangular endpoint theorem [5], Kalton's Hardy-type interpolation-stability theorem [6], Rutsky's real interpolation/AK-stability work [7], and the 2024 quasi-Banach function-space reiteration paper [4]. No full solution of the original bmo^c complex endpoint identity was found. Pisier and Kalton provide useful analogies but require analytic Hardy algebra or Köthe-lattice structure not available for the conditioned martingale subspaces.

References

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